
12-lead ECG image
(new site)
unchanged weights
Frozen ECG
multimodal LLM
PULSE-7B
MedGemma-4B
(no retraining)
(1) Generate an answer
“AF; RBBB”
Standard use (B0)
under-uses what the model knows
(2) Ask Yes/No per diagnosis
read P(yes) from next-token logits
Calibrate with n local labels
• threshold per label (very small n)
• S1: logistic regression on all
label logits (n ≈ 100–300)
(3) Read hidden state
(LLM layer 24–32, last token)
Linear probe on hidden state
(n ≥ 200–300)
Recommended
read-out
probability + threshold
S1
hidden-state probe
25
50
100
200
300
600
Local labelled ECGs available (n)